

Lab Manual: Building Your First Streamlit App

Objective

By the end of this lab, you will:

- Install and set up **Streamlit**.
- Build a simple interactive Python app.
- Add a title, descriptive text, user input, and tabular data.
- Run and view your app in a web browser.

Prerequisites

- Python 3.7 or higher installed.
- pip package manager available.
- (Optional) A virtual environment (recommended).

Step 1: Install Streamlit

Run the following command in your terminal: `pip install streamlit`

This installs Streamlit and makes it available in Python.

Step 2: Create a Python File

1. Open your code editor (VS Code, PyCharm, or any text editor).
2. Create a new file named: `app.py`

Step 3: Add a Title and Text

Paste the following code inside `app.py`:

```
import streamlit as st

# Add a title
st.title("Hello Streamlit!")

# Add descriptive text
st.write("This is my first Streamlit app.")
```

This creates a large heading and a descriptive line of text.

Step 4: Run the Streamlit App

In your terminal, navigate to the folder containing `app.py` and run:

```
streamlit run app.py
```

- This launches a local server.
- Your browser will automatically open <http://localhost:8501> to show the app.

Step 5: Add Interactivity

Update `app.py` with a text input box:

```
# Add a text input field
name = st.text_input("Enter your name:")

# Display a dynamic greeting if a name is entered
if name:
    st.write(f"Hello, {name}!")
```

Now the app responds dynamically to user input.

Step 6: Display Tabular Data

Add this code below the previous section:

```
import pandas as pd

# Create a sample DataFrame
data = pd.DataFrame({
    "Name": ["Alice", "Bob", "Charlie"],
    "Age": [25, 30, 35]
})

# Display the DataFrame in Streamlit
st.dataframe(data)
```

Streamlit renders it as an **interactive, scrollable table**.

Step 7: View the App

1. Save the file `app.py`.
2. Refresh your browser tab with the Streamlit app.
3. You should now see:
 - a. A title and description.
 - b. A text input box that greets users.
 - c. An interactive table with names and ages.

Verification Checklist

- Installed Streamlit successfully.
- Created `app.py` with title and text.
- App launched with `streamlit run app.py`.
- Added text input that responds dynamically.
- Displayed a DataFrame as an interactive table.

Summary

In this lab, you:

- Installed and set up **Streamlit**.
- Created a simple web app using only Python.
- Added **title, text, interactivity, and data display**.
- Ran the app locally in a browser.

With just a few lines of code, you've built a functional app! From here, you can expand with:

- Charts and plots (via `st.line_chart`, `st.bar_chart`)
- Maps (`st.map`)
- Machine learning model integration